

Real Algebra

Real Closed Fields and Galois-Theoretic Characterization

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Algebraic Background of Ordered Fields (Artin–Schreier)

Orderings and Preorderings of Fields

Definition (Ordering of a field)

Let K be a field. An **ordering** of K is a subset $P \subseteq K$ satisfying:

01 $P + P \subseteq P$ and $P \cdot P \subseteq P$

02 $P \cap (-P) = \{0\}$

03 $P \cup (-P) = K$

The pair (K, P) is an **ordered field**.

Notation: $P + P := \{a + b : a, b \in P\}$ and $P \cdot P := \{ab : a, b \in P\}$.

Remark on (O2)

If P is closed under addition and multiplication and satisfies (O3), then the condition (O2) is equivalent to

$$(O2') \quad -1 \notin P.$$

Algebraic Background of Ordered Fields

Orderings and Preorderings of Fields

Definition (Preordering of a Field)

A **preordering** of a field K is a subset $T \subseteq K$ satisfying:

- (P1) $T + T \subseteq T$ and $T \cdot T \subseteq T$
- (P2) $T \cap (-T) = \{0\}$
- (P3) $\forall a \in K, a^2 \in T$

Examples

- The set of non-negative reals $\mathbb{R}_{\geq 0}$ is the unique ordering of $\mathbb{R}$.
- Similarly, $\mathbb{Q}_{\geq 0}$ is the unique ordering of $\mathbb{Q}$.
- $\mathbb{C}$ *cannot* be ordered because -1 can be expressed as a sum of squares ($-1 = i^2$).

Algebraic Background of Ordered Fields

Properties of Preorderings & Sums of Squares

Remarks on Preorderings

- Similarly, assuming (P1) and (P3), condition (P2) is equivalent to **(P2')** $-1 \notin T$.
- Every ordering is a preordering. Indeed, (P3) is a weaker version of (O3), where $\forall a \in K, (a \in P \vee -a \in P)$ implies $a^2 \in P$ by (O1).
- Arbitrary intersections of preorderings are again preorderings.

The Smallest Preordering

We denote the set of sums of squares in K by $\sum K^2$:

$$\sum K^2 := \{a_1^2 + \cdots + a_n^2 \mid n \in \mathbb{N}, a_1, \dots, a_n \in K\}$$

$\sum K^2$ is contained in every preordering of K . And It is a preordering if and only if $-1 \notin \sum K^2$.

Algebraic Background of Ordered Fields

Real Fields

Definition (Definition 1.1.7: Real Field)

The field K is called **real** if $-1 \notin \sum K^2$, i.e., when -1 cannot be written as a sum of squares in K .

Immediate Consequences

- A field K has a preordering **if and only if** it is real.
- In this case, $\sum K^2$ is the *smallest* preordering on K .
- All real fields must have characteristic 0.

Ref: M. Knebusch & C. Scheiderer, *Real Algebra. A First Course* (2022)

Examples